Warm up by hand

W1: $P(S) = 6/20 = 0.3$

$P(H) = 4/20 = 0.2$

$P(Q) = 5/20 = 0.25$

W2: Conditional

find $P(S|H)$ and $P(S|H^c)$

$P(S|H) = 3/4$

$P(S|H^c) = 3/16$

$3/4 \div 3/16 = 4$

Strong week is ~~more likely in~~ a 4 times more likely in a holiday week than in a not a holiday week.

W3: $P(H|S) = \frac{P(S|H)\,P(H)}{P(S)}$

$= \frac{3/4 \times 0.2}{0.3}$

$= 3/4 \times 2/3 = \boxed{0.5}$ Bayes theorem

$P(H|S) = 3/6 = 0.5$ by counting

W4

$$P(S) = P(S|H)P(H) + P(S|H^c)P(H^c)$$

$$= \frac{3}{4} \times \frac{1}{5} + \frac{3}{16} \times \frac{16}{20}$$

$$= \frac{3}{20} + \frac{3}{20}$$

$$= \frac{6}{20} = 0.3$$

W5 Is $P(S \cap H) = P(S)P(H)$?

$P(S \cap H) = 3/20 = 0.15$

$P(S)P(H) = 6/20 \times 4/20 = 0.3 \times 0.2 = 0.06$

Since $0.15 \neq 0.06$, S and H are not independent. In plain words, knowing its holiday week changes our belief about whether it will be a strong week. If S and H were independent $P(S|H)$ would equal $P(S) = 0.3$. But $P(S|H) = 0.75$. So, both events are dependent.

W6 Precision $P(S|H)$ | Recall $= P(H|S)$

$P(S|H) = 0.75$ | $= 0.5$

$1 - \text{recall} = 0.5$, so $1/2$ of the strong weeks are missed

W7

→ $P(S)\,P(H|S)\,P(Q|S)$
$= 0.3 \times 0.5 \times 4/6$
$= 0.15 \times 4/6$
$= 3/20 \times 2/3$
$= 0.1$

→ $P(S^c)\,P(H|S^c)\,P(Q|S^c)$
$= 0.7 \times 1/14 \times 1/14$
$= 0.00357$

$P(S|x) = P(S) \cdot P(x|S) / P(x)$

→ Normalization

$P(S|H,Q) = 0.1 / (0.1 + 0.00357) \approx 0.966$
$P(S'|H,Q) = 0.00357 / (0.1 + 0.00357) \approx 0.034$

Check $(R\bar{S}) = 0.966 + 0.034 = 1$

→ Score for $P(S)\,P(H^c|S)\,P(Q^c|S)$ and $P(S^c)P(H^c|S^c)P(Q^c|S^c)$

→ $P(S)\,P(H^c|S)\,P(Q^c|S)$
$= 0.3 \cdot 0.5 \cdot 1/3$
$= 3/10 \times 1/2 \times 1/3$
$= 0.05$

→ $P(S^c)\,P(H^c|S^c)\,P(Q^c|S^c)$
$= 0.7 \cdot 13/14 \cdot 13/14$
$= 0.6$

~~Normalize $0.2/(0.2+0.6)$~~
~~$P(S|H^c,Q^c) = 0.25$~~

$0.6/(0.2+0.6)$
~~$P(S^c|H^c,Q^c) = 0.75$~~

→ Normalization

$P(S|H^c,Q^c) = 0.05/(0.05+0.6) = 0.077$

$P(S^c|H^c,Q^c) = 0.6/(0.05+0.6) = 0.923$

Scanned with CamScanner

Probability from the column of ~~P(S)~~ P(S | H, Q)

$= \frac{2}{2+0} = 1$

Through naive bayes, 0.966, close!

Probability from the column of $P(S^c | H^c, Q^c)$

$\frac{1}{1+12} = \frac{1}{13} = 0.0769 \approx 0.77$

Through naive bayes, 0.077, close again!